BRAC University

Department of Mathematics and Natural Sciences

Lecture on

Real Analysis II (MAT321)

Uniform Convergence

Poinwise and Uniform Convergence, Cauchy Criterion and Test for Uniform Convergence

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Conducted by

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Convergence of Sequence in Real Numbers

Convergence of a Sequence

A sequence $\{a_n\}$ is said to **converge** to a real number a if for every real number $\varepsilon > 0$, there exists a natural number $N \in \mathbb{N}$ such that for all $n \geq N$, the terms of the sequence satisfy the inequality

$$|a_n - a| < \varepsilon.$$

If such a number a exists, it is called the **limit** of the sequence, and we write

$$\lim a_n = a \quad \text{or} \quad a_n \to a \text{ as } n \to \infty.$$

Cauchy Criterion for Convergence

A sequence $\{a_n\}$ is said to satisfy the **Cauchy criterion** if for every real number $\varepsilon > 0$, there exists a natural number $N \in \mathbb{N}$ such that for all $m, n \geq N$, the terms of the sequence satisfy the inequality

$$|a_n - a_m| < \varepsilon.$$

Note: A sequence that satisfies the Cauchy criterion is called a **Cauchy sequence**. In the real numbers, every Cauchy sequence converges to a limit, and every convergent sequence is a Cauchy sequence.

Limit and Continuity of a function

Limit of a Function

Let $f : A \to \mathbb{R}$ be a function defined on a subset A of the real numbers, and let c be a limit point of A. We say that the **limit** of $f(x)$ as x approaches c is equal to L, and we write

$$\lim_{x \to c} f(x) = L,$$

if for every $\varepsilon > 0$, there exists a $\delta > 0$ such that for all $x \in A$ with $0 < |x - c| < \delta$, we have

$$|f(x) - L| < \varepsilon.$$

Continuity of a Function

A function $f : A \to \mathbb{R}$ is said to be **continuous** at a point $c \in A$ if

$$\lim_{x \to c} f(x) = f(c).$$

In other words, f is continuous at c if for every $\varepsilon > 0$, there exists a $\delta > 0$ such that for all $x \in A$ with $|x - c| < \delta$, we have

$$|f(x) - f(c)| < \varepsilon.$$

Sequence and Series of Functions and their Convergence

Sequence of Functions

A **sequence of functions** is a collection of functions $\{f_n\}$ where each function f_n is defined on a common domain D. Each function in the sequence is indexed by a natural number n, and we can write the sequence as

$$f_1, f_2, f_3, \ldots, f_n, \ldots$$

Series of Functions

A **series of functions** is an infinite sum of functions, typically expressed as

$$\sum_{n=1}^{\infty} f_n(x),$$

where each f_n is a function defined on a common domain D. The series converges at a point x if the sequence of partial sums

$$S_N(x) = \sum_{n=1}^{N} f_n(x)$$

converges as $N \to \infty$.

Poinwise Convergence

Pointwise Convergence of a Sequence of Functions

A sequence of functions $\{f_n\}$ defined on a domain D is said to **converge pointwise** to a function f on D if for **every** $x \in D$ and for every $\varepsilon > 0$, there exists a natural number $N \in \mathbb{N}$ (which may depend on x and ε) such that for all $n \geq N$, the following inequality holds:

$$|f_n(x) - f(x)| < \varepsilon.$$

In this case, we write

$$\lim_{n\to\infty} f_n(x) = f(x) \quad \text{for all } x \in D.$$

Pointwise Convergence of Series of Functions

A series of functions $\sum_{n=1}^{\infty} f_n$ defined on a domain D is said to **converge pointwise** to a function F on D if for **every** $x \in D$ the sequence of partial sums

$$S_N(x) = \sum_{n=1}^{N} f_n(x)$$

converges pointwise to $F(x)$ as $N \to \infty$.

Some Problems on Pointwise Convergence

Pointwise Limit of Continuous Functions

Define a sequence of functions $\{f_n\}$ on $[0, 1]$ by

$$f_n(x) = x^n.$$

1. Find the pointwise limit $f(x)$ of $\{f_n\}$ on $[0, 1]$.
2. Show that each f_n is continuous on $[0, 1]$.
3. Show that the limit function f is not continuous on $[0, 1]$.

Pointwise Convergent Series with Discontinuous Sum

For $n \in \mathbb{N}$, define $f_n : \mathbb{R} \to \mathbb{R}$ by

$$f_n(x) = \frac{x^2}{(1+x^2)^n}.$$

Consider the series of functions

$$\sum_{n=0}^{\infty} f_n(x).$$

1. Show that each f_n is continuous on $[0,1]$.
2. Show that for each $x \in [0,1)$, the series converges and find its sum.
3. Show that the sum function is not continuous on $[0,1]$.

Pointwise Limit of Differentiable Functions

Define $f_n : [-1, 1] \to \mathbb{R}$ by

$$f_n(x) = x^{1+\frac{1}{2n-1}}.$$

1. Show that $f_n(x) \to |x|$ pointwise on $[-1, 1]$.
2. Show that each f_n is differentiable on $[-1, 1]$.
3. Show that the limit function $f(x) = |x|$ is not differentiable at $x = 0$.

Interchanging Limit and Derivative

Define $f_n : \mathbb{R} \to \mathbb{R}$ by

$$f_n(x) = \frac{\sin(nx)}{\sqrt{n}}.$$

1. Show that $f_n(x) \to 0$ for all $x \in \mathbb{R}$.
2. Compute $f_n'(x)$.
3. Conclude that

$$\left(\lim_{n\to\infty} f_n\right)' \neq \lim_{n\to\infty} f_n'.$$

Interchanging Limit and Integral

Define $f_n : [0,1] \to \mathbb{R}$ by

$$f_n(x) = nx(1-x^2)^n.$$

1. Show that $f_n(x) \to 0$ for every $x \in [0,1]$.
2. Compute $\int_0^1 f_n(x)\,dx$.
3. Conclude that

$$\lim_{n\to\infty} \int_0^1 f_n(x)\,dx \neq \int_0^1 \lim_{n\to\infty} f_n(x)\,dx.$$

Interchanging Sum and Integral

Define $f_n : [0,1] \to \mathbb{R}$ by

$$f_n(x) = \begin{cases} n^2 x, & 0 \le x \le \frac{1}{n}, \\ n^2\left(\frac{2}{n} - x\right), & \frac{1}{n} < x \le \frac{2}{n}, \\ 0, & \frac{2}{n} < x \le 1. \end{cases}$$

Show that $\sum f_n(x)$ converges pointwise on $[0,1]$, but

$$\int_0^1 \sum_{n=1}^{\infty} f_n(x)\, dx \neq \sum_{n=1}^{\infty} \int_0^1 f_n(x)\, dx.$$

Uniform Convergence

Uniform Convergence of a Sequence of Functions

A sequence of functions $\{f_n\}$ defined on a domain D is said to **converge uniformly** to a function f on D if for every $\varepsilon > 0$, there exists a natural number $N \in \mathbb{N}$ such that for all $n \geq N$ and for all $x \in D$, the following inequality holds:

$$|f_n(x) - f(x)| < \varepsilon.$$

In this case, we write

$$f_n \to f \text{ uniformly on } D.$$

Note: In uniform convergence, the choice of N depends only on ε and not on x, which is a stronger condition than pointwise convergence.

Uniform Convergence of Series of Functions

A series of functions $\sum_{n=1}^{\infty} f_n$ defined on a domain D is said to **converge uniformly** to a function F on D if the sequence of partial sums

$$S_N(x) = \sum_{n=1}^{N} f_n(x)$$

converges uniformly to $F(x)$ as $N \to \infty$.

Problem

Test whether the sequence of functions $\{f_n\}$ defined by

$$f_n(x) = \frac{\sin(nx)}{\sqrt{n}}$$

converges uniformly on the interval $[0, 2\pi]$.

Problem

Test whether the sequence of functions $\{f_n\}$ defined by

$$f_n(x) = \frac{x}{1 + nx^2}$$

converges uniformly on the interval $[0, 1]$.

Problem

Show that the sequence of functions $\{f_n\}$ defined by

$$f_n(x) = x^n$$

is uniformly convergent on $[0, k]$ for every $0 < k < 1$, but converges only pointwise on $[0, 1]$.

Problem

Show that the sequence of functions $\{f_n\}$ defined by

$$f_n(x) = \frac{1}{x+n}$$

is uniformly convergent on any interval $[0, b]$, where $b > 0$.

Problem

Show that $f_n(x) = \tan^{-1}(nx)$ is uniformly convergent on $[a, b]$ for $a > 0$, but not uniformly convergent on $[0, b]$.

Cauchy Criterion for Uniform Convergence

Cauchy Criterion for Uniform Convergence

A sequence of functions $\{f_n\}$ defined on a domain D is said to satisfy the **Cauchy criterion for uniform convergence** if for every $\varepsilon > 0$, there exists a natural number $N \in \mathbb{N}$ such that for all $m, n \geq N$ and for all $x \in D$, the following inequality holds:

$$|f_n(x) - f_m(x)| < \varepsilon.$$

Cauchy Criterion for Uniform Convergence of Series

A series of functions $\sum_{n=1}^{\infty} f_n$ defined on a domain D is said to satisfy the **Cauchy criterion for uniform convergence** if for every $\varepsilon > 0$, there exists a natural number $N \in \mathbb{N}$ such that for all $m > n \geq N$ and for all $x \in D$, the following inequality holds:

$$|f_{n+1}(x) + f_{n+2}(x) + f_{n+3}(x) + \cdots + f_m(x)| < \varepsilon.$$

Test for Uniform Convergence of Sequence of Functions

Uniform Convergence via Supremum Norm

Let $\{f_n\}$ be a sequence of functions defined on a set D, and let $f : D \to \mathbb{R}$ be a function. For each n, define

$$M_n = \sup_{x \in D} |f_n(x) - f(x)|.$$

Then $f_n \to f$ uniformly on D if and only if

$$M_n \xrightarrow[n\to\infty]{} 0.$$

Remark

The assumption that $f_n \to f$ pointwise is not needed: if $M_n \to 0$, then $f_n \to f$ automatically (indeed, uniformly).

Problem

Consider the sequence of functions $f_n : \mathbb{R} \to \mathbb{R}$, $n \geq 1$, defined by

$$f_n(x) = \frac{x}{nx^2 + 1}, \qquad x \in \mathbb{R}.$$

Does $\{f_n\}$ have a uniform limit on $\mathbb{R}$? Justify your answer.

Problem

Consider the sequence of functions $f_n : (0,1) \to \mathbb{R}$, $n \geq 1$, defined by

$$f_n(x) = \frac{nx}{nx^2 + 1}, \qquad 0 < x < 1.$$

Does $\{f_n\}$ have a uniform limit on $(0,1)$? Justify your answer.

Problem

Show that the sequence of functions $\{f_n\}$ defined on $[0,1]$ by

$$f_n(x) = \begin{cases} nx, & 0 \le x \le \dfrac{1}{n}, \\ \dfrac{1}{n}, & \dfrac{1}{n} < x \le 1, \end{cases}$$

is not uniformly convergent on $[0,1]$.

Problem

Let $\{f_n\}$ be a sequence of functions defined on $[0, \infty)$ by

$$f_n(x) = nxe^{-nx^2}, \qquad x \geq 0.$$

Does $\{f_n\}$ have a uniform limit on $[0, \infty)$? Justify your answer.

Test for Uniform Convergence of Series of Functions

Weierstrass M-Test for Series of Functions

Let $\{f_n\}$ be a sequence of functions defined on a set D. Assume there exists a sequence of real numbers $\{M_n\}$ with $M_n \geq 0$ such that

$$|f_n(x)| \leq M_n \quad \text{for all } x \in D \text{ and all } n.$$

If the numerical series $\sum_{n=1}^{\infty} M_n$ converges, then the function series

$$\sum_{n=1}^{\infty} f_n(x)$$

converges uniformly on D (also absolutely on D).

Remark

The converse is false in general: uniform convergence of $\sum f_n$ does not imply that $\sum \sup_{x \in D} |f_n(x)|$ converges.

Problem

Show that the series of functions

$$\sum_{n=1}^{\infty} \frac{\sin(x^2 + n^2 x)}{n(n+1)}$$

is uniformly convergent for all real numbers x.

Thank You!

We'd love your questions and feedback.

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References

[1] Stephen Abbott, *Understanding Analysis*, 2nd Edition, Springer, 2015.

[2] Walter Rudin, *Principles of Mathematical Analysis*, 3rd Edition, McGraw-Hill, 1976.